

Name: Tyler Eaton

675484223

55

QUIZ 3

TRANSPORT PHENOMENA II

CLOSED BOOK, TIME ALLOWED: 40 MIN

Consider a porous cylindrical catalyst with radius $r=R$ and length $z=L$. Species A diffuses into the catalyst. The zeroth-order reaction rate coefficient is k_0'' (moles of A generated per unit time per unit area is $-k_0''$). At the catalyst surface, concentration of A is specified as $c_A(R) = c_{A0}$. Assume:

- Stagnant system (neglect convection),
- Diffusion only happens in the radial direction,
- Steady state & isothermal conditions.

$$\frac{dN_A}{dt} = 0$$

$$c_A = c_A(r)$$

$$c_A \langle v \rangle^* = 0$$



- 1- Given that the mole fraction of A in the catalyst is very small ($x_A \ll 1$), simplify the combined mole flux

$$\langle N_A \rangle = c_A \langle v_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$$

to

$$\langle N_A \rangle = -D_A \langle \nabla c_A \rangle$$

Recall: $\langle J_A^* \rangle = -c D_A \langle \nabla x_A \rangle$ where c is the total concentration.

- 2- Set up a shell balance to derive an ODE for the steady-state concentration profile of A as a function of r

$$\int_V dV \frac{\partial c_A}{\partial t} = - \int_S dS \langle N_A \rangle_n + \int_V dV R_A$$

- 3- List 2 boundary conditions.

- 4- Solve the ODE and determine the integration constants.

Your answers may be in terms of the following constants:

R and L

k_0'' : surface reaction rate constant in moles per unit time per unit area

a : available surface area per unit volume of the catalyst.

D_A : effective diffusivity of species A in the porous medium, where Fick's law in terms of effective diffusivity is

c_{A0} : molar concentration of A at the surface of the catalyst.

$$\langle N_A \rangle = c_A \langle v_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$$

$$\langle J_A^* \rangle = -D_A \langle \nabla c_A \rangle$$

$$0 = -S \langle \nabla c_A \rangle + \int_V dV R_A$$

$$\langle N_A \rangle = c_A \langle v_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$$

$$\langle N_A \rangle = c_A \langle v^* \rangle - D_A \langle \nabla c_A \rangle$$

$$\langle N_A \rangle = c_A \langle v_A \rangle = -D_A \langle \nabla c_A \rangle$$

$$= -D_A \langle \nabla c_A \rangle$$

$$\frac{d}{dt} \int_V dV c_A = - \int_S dS \langle N_A \rangle_n + \int_V dV R_A$$

$$0 = - \int_S dS \langle N_A \rangle_n + \int_V dV R_A$$

$$= - \int_S dS \langle N_A \rangle_n + \int_V dV R_A$$

$$= 2\pi R \langle N_A \rangle_n - 2\pi R \langle N_A \rangle_n$$

2)

$$2\pi r < -D_A < \frac{D_A}{r} - 2\pi R < -D_A < \frac{D_A}{r} \ln 7$$

$$-\int_r^R ds < -D_A < \frac{D_A}{r} \ln 7 \quad S_A = 2\pi r$$

3) B.c.

$$\begin{aligned} r=R \\ C_A = C_{A0} \\ dC_A = 0 \\ r=r? \end{aligned}$$

$$-\int_r^R 2\pi r dr < -D_A < \frac{D_A}{r} \ln 7$$

$$2\pi D_A \int_r^R \frac{dr}{r} < \frac{D_A}{r} \ln 7$$

$$2\pi D_A \int_r^R \frac{dr}{r} < \frac{C_A(r)}{r} \ln 7$$

$$\left(2\pi r D_A < \frac{C_A(r)}{r} \ln 7 \right)_{r+\Delta r} - \left(2\pi r D_A < \frac{C_A(r)}{r} \ln 7 \right)_r = -K''$$

$$\left(r < \frac{C_A(r)}{r} \ln 7 \right)_{r+\Delta r} - \left(r < \frac{C_A(r)}{r} \ln 7 \right)_r = -\frac{K''}{2\pi D_A}$$

$$r=R$$

$$C_A = C_{A0}$$

$$2\pi R D_A C_{A0} = \int_V \frac{dr^2}{dt} C_A \pi$$

$$2\pi R D_A C_{A0} = 2\pi R C_{A0} \pi R$$

$$t D_A = 1 + C_1$$

$$r=r$$

$$dC_A = 0$$

$$2\pi r D_A C_A = \int \frac{dr^2}{dt} C_A \pi$$

$$2\pi r D_A C_A t = 2\pi r \frac{dC_A(r)}{dt} \pi + C_2$$

$$2\pi r D_A C_A t = C_2$$

$$C_1 = t D_A - 1$$

$$C_2 = 2\pi r D_A C_A t$$

$$\begin{aligned} \int_V dv \frac{dC_A}{dt} &= \int_V \frac{dv}{dt} C_A + \frac{dC_A}{dt} V \\ dC_A \cdot V &= dV_A \\ \int_V \frac{dv}{dt} C_A &= K'' \\ \frac{dC_A}{dt} &= 0 \end{aligned}$$